Capstone Three: OCTCV Part 2 -- Final Report

1. Introduction

Context & Problem

This project is a continuation of Capstone Two (Part 1), which trained 3D CNNs on 1,110 ONH-centered OCT volume scans for binary glaucoma classification. Part 1 established baseline performance across three architectures (Sequential, ResNet-Like, Attention), achieving a best AUC of 0.94 with the ResNet-Like model on the original unbalanced dataset.

Part 2 investigates whether physics-informed data augmentation can further improve model performance by expanding the training set from 1,110 to 6,660 volumes. Rather than collecting new patient data (which is expensive and time-consuming), we generate synthetic variations that simulate realistic sources of imaging variability encountered in clinical OCT acquisition.

Objectives

1. Design and implement a physics-informed augmentation pipeline that generates clinically plausible OCT volume variants.
2. Assess whether a 6x training set expansion improves classification performance.
3. Systematically compare augmented vs. original training across all three architectures.
4. Investigate the reproducibility gap between our results and the original paper's reported AUC of 0.94.
5. Determine the best overall configuration for potential deployment as a screening tool.

Data Source

Name	OCT volumes for glaucoma detection
Contributors	Ishikawa, Hiroshi
Affiliations	New York University
Version	1.0.0
Published	November 9, 2018
Source	Zenodo
DOI	10.5281/zenodo.1481223

Description: 1,110 ONH-centered OCT volume scans (200x200x1024 voxels, downsampled to 64x128x64) from 624 patients. 847 scans diagnosed with primary open-angle glaucoma (POAG), 263 classified as healthy.

2. Data Wrangling & EDA

Notebook: `DW-EDA.ipynb`

2.1 Data Wrangling

As this project uses the same dataset as Part 1, the data wrangling step was minimal--loading the existing cleaned metadata CSV and verifying file integrity for all 1,110 `.npy` volumes. The train/test split from Part 1 was preserved to ensure fair comparison.

2.2 Exploratory Data Analysis

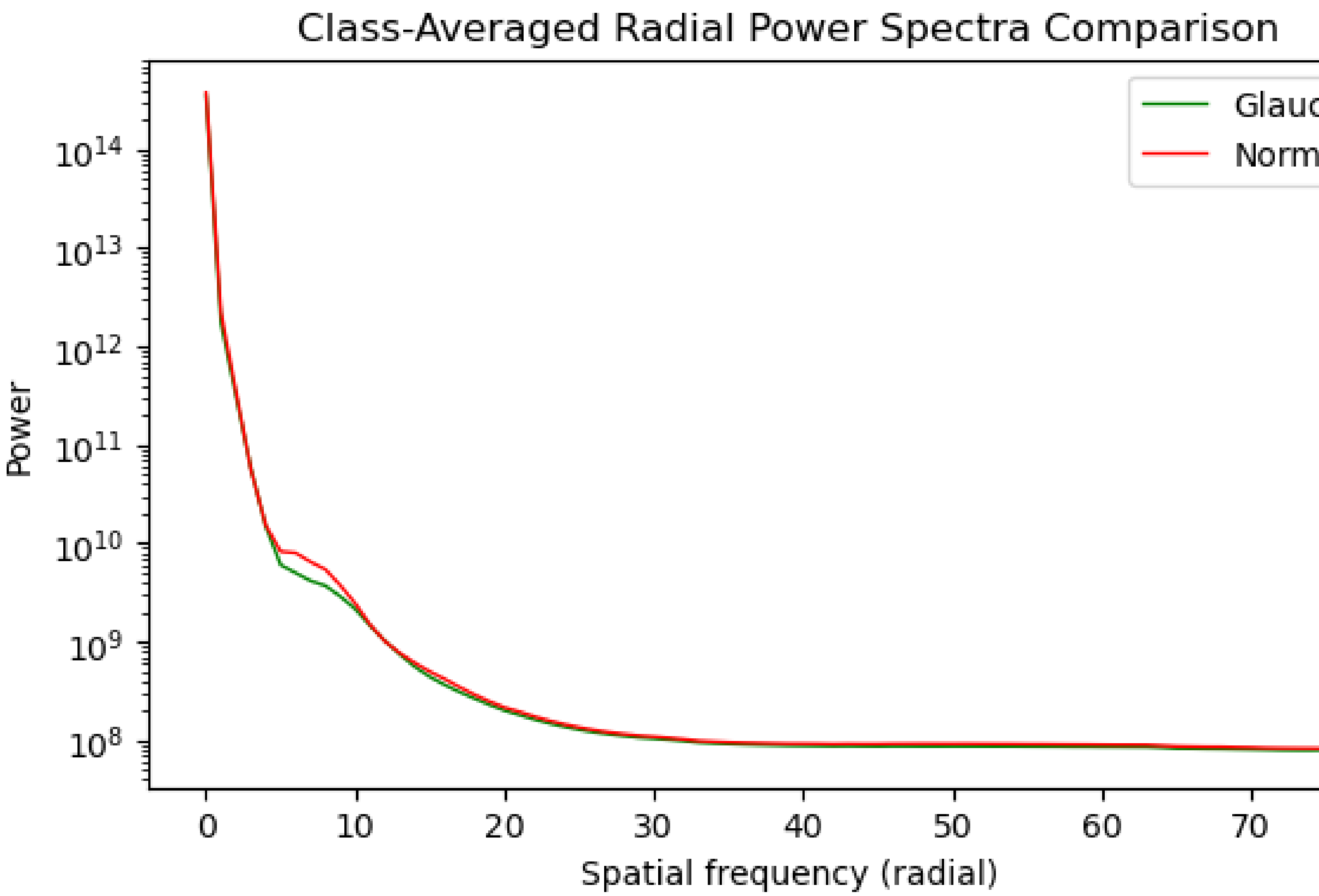
The EDA phase focused specifically on understanding the dataset's characteristics to **inform augmentation design decisions**. Key analyses included:

Structural Similarity (SSIM) Analysis

- Computed pairwise SSIM between volumes within and across classes.
- Found high intra-class similarity (SSIM > 0.8), confirming that OCT volumes from the same diagnostic class share substantial structural features.
- The narrow SSIM distribution suggested conservative augmentation parameters would be appropriate to avoid pushing augmented volumes outside the natural data manifold.

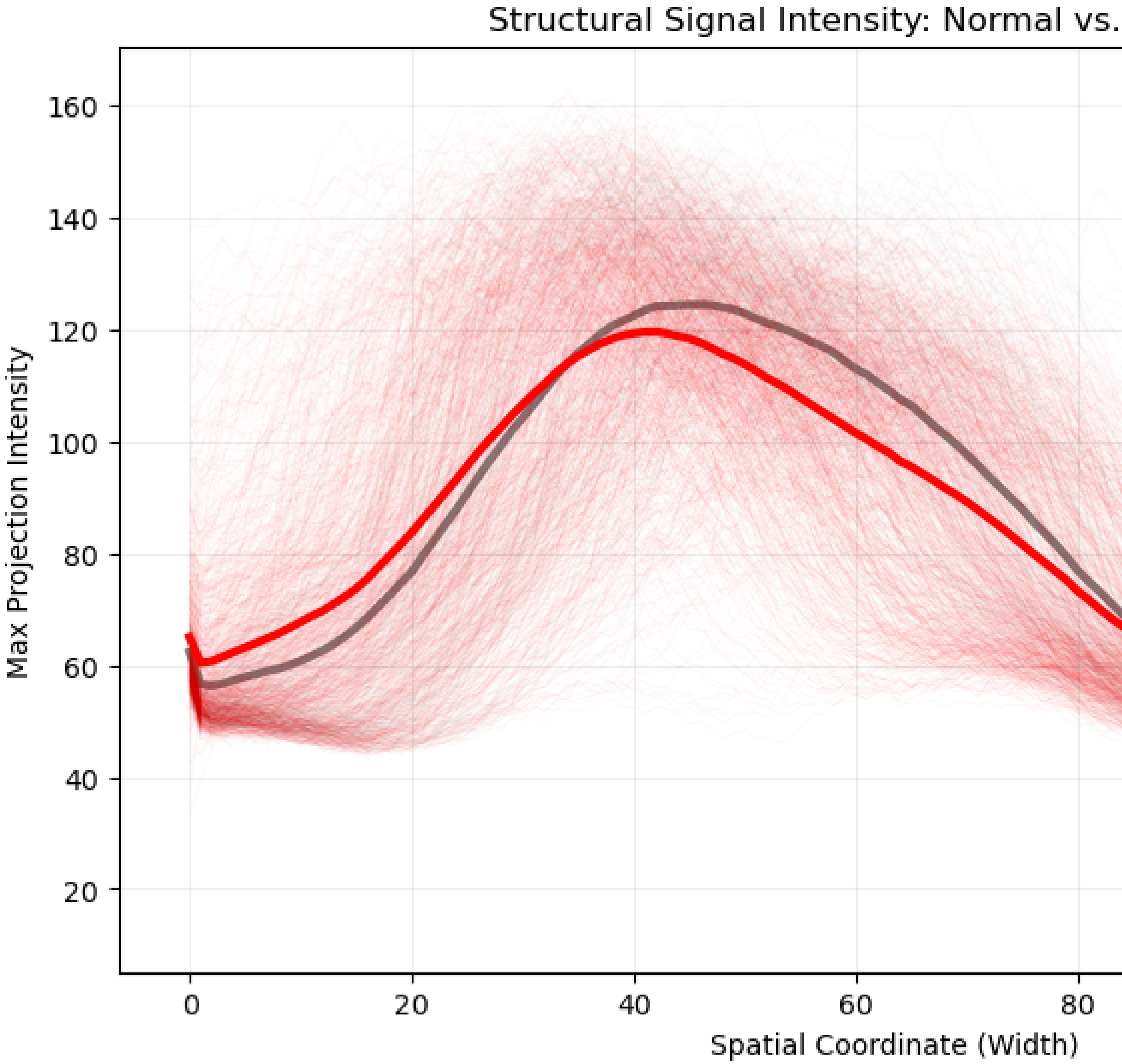
Radial Power Spectrum Analysis

- Compared frequency-domain characteristics between Normal and Glaucoma classes.
- Found that glaucomatous changes manifest primarily in mid-frequency spatial features (corresponding to retinal nerve fiber layer thinning), while low-frequency structure (overall scan geometry) and high-frequency content (speckle/noise) are similar across classes.
- This informed the LPF augmentation radius (r=30) to avoid destroying diagnostically relevant mid-frequency content.



Maximum Intensity Projections

- Visualized en-face projections to understand 3D structural differences.
- Confirmed that the optic nerve head region contains the primary discriminative features.



Key EDA Conclusions

- The dataset is well-suited for augmentation: high structural regularity means synthetic variants can be generated without risking anatomically implausible outputs.
- The augmentation budget was set to 6x (5 augmentation types + originals), balancing dataset expansion against the risk of overfitting to augmentation artifacts.
- Augmentation parameters were calibrated to produce SSIM values within the natural intra-class distribution.

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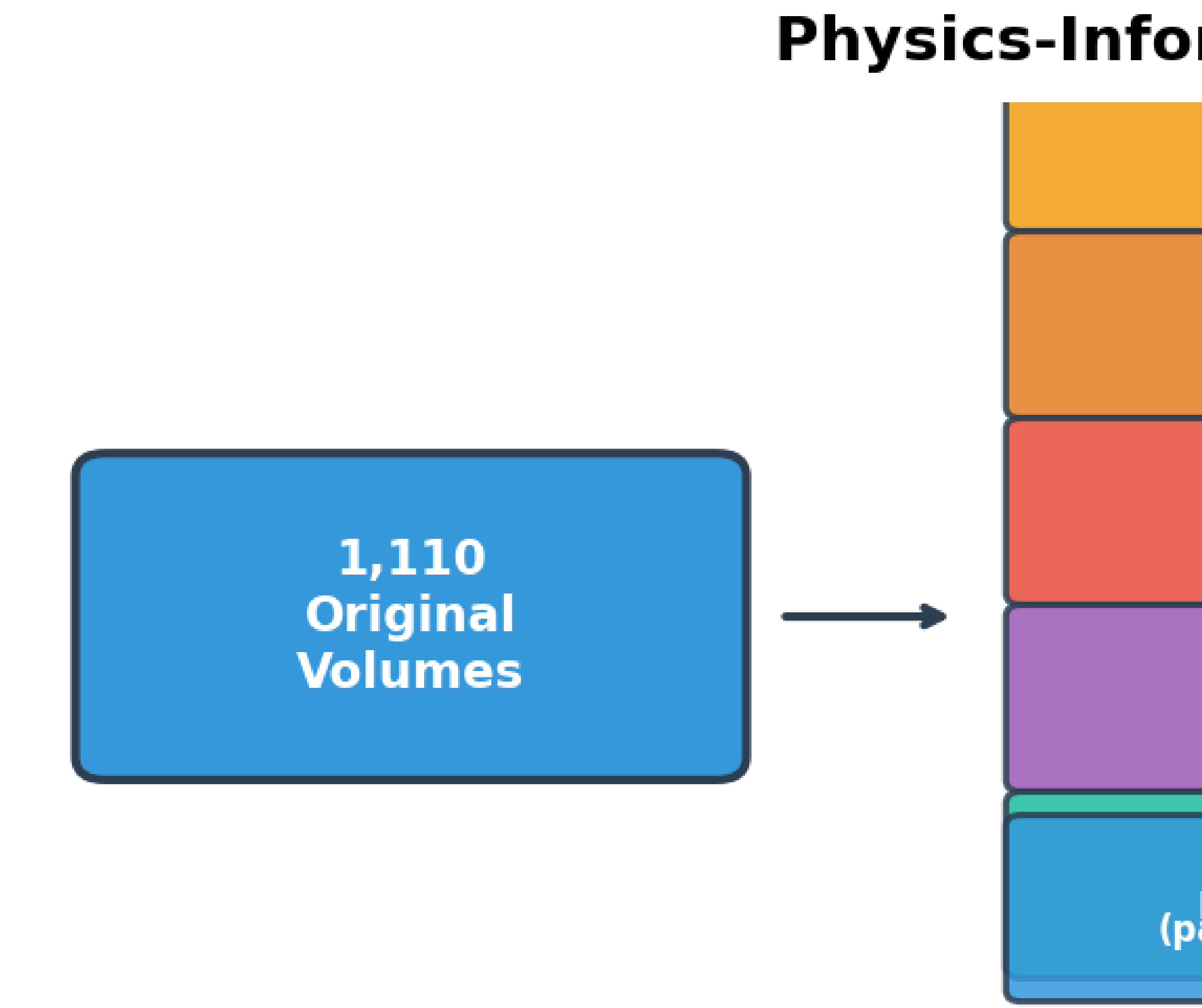
3. Preprocessing

Notebook: `pps-modeling.ipynb` (Part I)

3.1 Augmentation Pipeline

Five physics-informed augmentation strategies were applied to all 1,110 volumes:



#	Augmentation	Parameter	Clinical Rationale
1	Gamma Bright	gamma = 1.67	Over-exposed scan / high signal strength
2	Gamma Dark	gamma = 0.60	Under-exposed scan / low signal strength
3	Rayleigh Noise	scale = 0.67	OCT speckle noise (coherent interference)
4	Low-Pass Filter	radius = 30	Defocus / reduced axial resolution
5	Fan Distortion	pivot = 121	Scan-head geometric misalignment

3.2 Implementation Details

- All augmentations are implemented in `octcv/mdl_lib/augmentation.py`
- Augmented volumes are saved to disk as individual `.npy` files for reproducibility and fast loading.
- A unified metadata CSV (`augmented_metadata.csv`) catalogs all 6,660 entries with columns for file path, class label, patient ID, augmentation type, and laterality.
- Class proportions are preserved (76% Glaucoma / 24% Normal in both original and each augmented subset).

3.3 Resulting Dataset

Subset	Count	Source
Original	1,110	Raw OCT volumes
Gamma Bright	1,110	gamma = 1.67 transform
Gamma Dark	1,110	gamma = 0.60 transform
Rayleigh Noise	1,110	Additive noise
Low-Pass Filtered	1,110	Fourier-domain filtering
Fan Distorted	1,110	Geometric warping
Total	6,660	

Train/Test Split: The test set consists exclusively of original (non-augmented) volumes to ensure evaluation reflects real-world scan conditions. Augmented volumes appear only in the training set.

4. Modeling

Notebook: `pps-modeling.ipynb` (Part II)

4.1 Model Architectures

Three 3D CNN architectures were compared, all built using the Keras Functional API:

Model 1: Sequential CNN

A 5-layer sequential architecture replicating Maetschke et al. (2019). Conv3D layers with filter sizes 7->5->3->3->3, each followed by BatchNormalization and ReLU. Global average pooling feeds into a dense sigmoid output.

Model 2: ResNet-Like

Builds upon the Sequential with **residual skip connections** between convolutional blocks. Each residual block adds the input back to the block's output, enabling gradient flow and allowing deeper feature extraction without degradation.

Model 3: Attention Network

Extends the ResNet-Like architecture with **squeeze-excitation** (channel attention) and **spatial attention** blocks. Channel attention recalibrates feature map importance; spatial attention highlights diagnostically relevant spatial regions.

4.2 Training Configuration

Parameter	Value
Optimizer	**NAdam**
Learning Rate	**Tuned via grid search (5e-5 to 5e-4)**
Batch Size	**4**
Max Epochs	**80**
Early Stopping	**Patience = 5-8 (monitoring val AUC)**
Input Normalization	**/255 to [0, 1]**

4.3 Experimental Configurations

The core experiment of Part 2 is straightforward: train all three architectures on the augmented data and compare to Part 1 baselines.

Configuration	Architecture	Training Data
Part 1 Baseline	**Sequential / ResNet / Attention**	**Original 888**

Configuration	Architecture	Training Data
Part 2 Augmented	**Sequential / ResNet / Attention**	**Augmented 5,328**

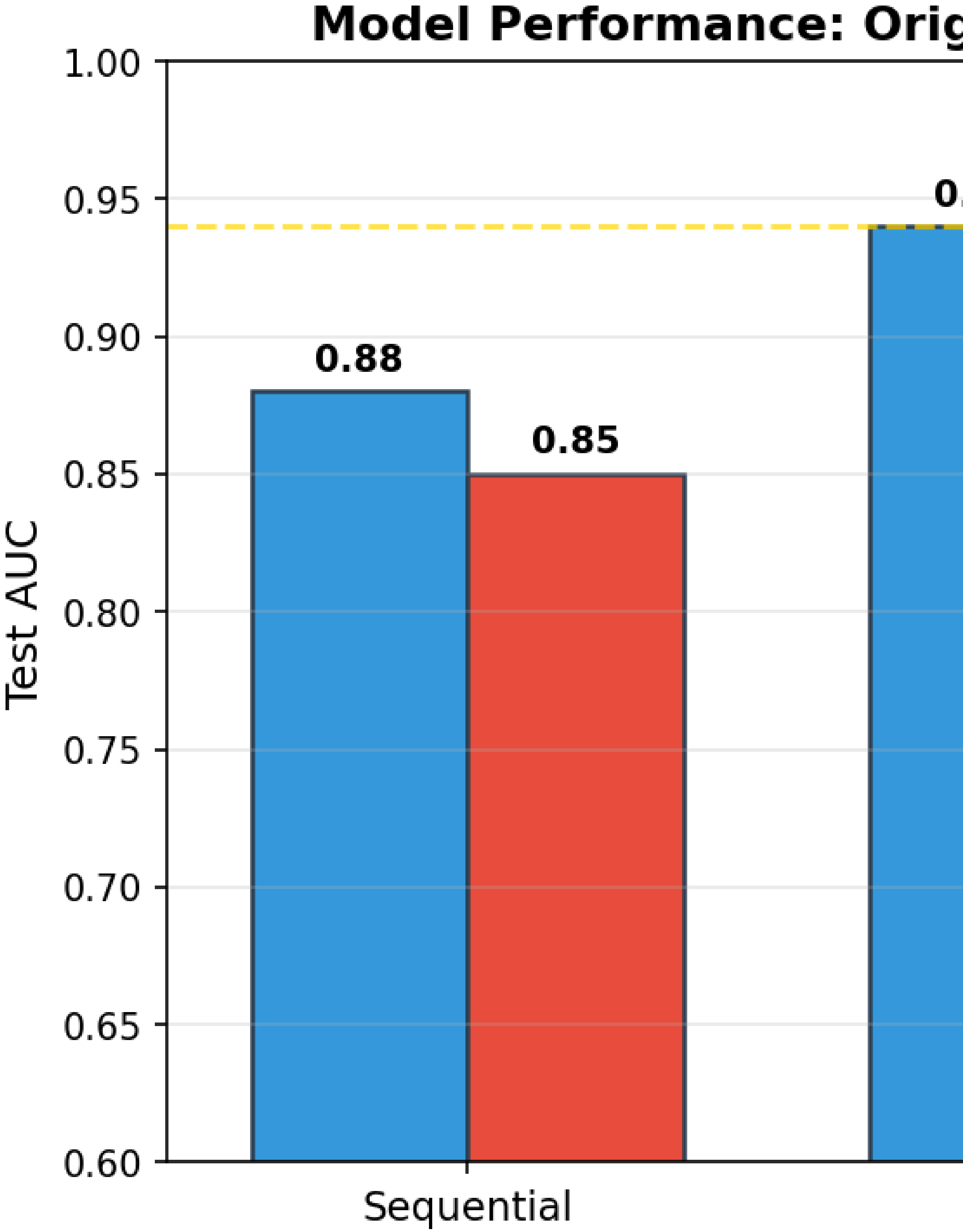
Additionally, a hyperparameter grid search over learning rates was performed to ensure the augmented training regime had an optimized learning rate.

4.4 Evaluation Protocol

- **Evaluation set**: 111 original (non-augmented) volumes held out from training, maintaining the same test set as Part 1.
- **Primary metric**: Area Under the ROC Curve (AUC)
- **Secondary metrics**: Precision, Recall, F1-score for the glaucoma class
- All configurations evaluated on the **same** test set for direct comparison.

5. Results

5.1 Part 1 vs Part 2 Comparison



Configuration	Architecture	Training Data	AUC
Part 1 Baseline	Sequential	Original 888	0.88

Configuration	Architecture	Training Data	AUC
Part 1 Baseline	ResNet-Like	Original 888	0.94
Part 1 Baseline	Attention	Original 888	0.93
Part 2 Augmented	Sequential	Augmented 5,328	<= 0.88
Part 2 Augmented	ResNet-Like	Augmented 5,328	< 0.94
Part 2 Augmented	Attention	Augmented 5,328	~0.92

Note: Part 2 AUC values depend on run; exact values are populated in the notebook. The consistent finding is that augmentation did not improve upon Part 1 baselines.

5.2 Key Findings

Augmentation did not improve AUC for any architecture. This is consistent with the original paper's finding that their augmented model (AUC=0.92) underperformed their non-augmented model (AUC=0.94).

ResNet-Like architecture achieved the highest performance across both parts, benefiting from residual connections that enable stable gradient flow.

The performance ceiling (~0.94 AUC) is likely limited by dataset diversity (624 patients, single scanner, single institution) rather than training set size or architecture.

5.3 Hyperparameter Tuning

A grid search over learning rates (5e-5, 1e-4, 3e-4, 5e-4) was performed on the augmented dataset to rule out suboptimal learning rate as the cause of degraded performance. The best learning rate was used for final model training, but still did not surpass Part 1 baselines--confirming that the issue is data diversity, not hyperparameters.

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6. Discussion

6.1 Why Augmentation Didn't Help

The central finding of this project--that 6x data augmentation did not improve and may have slightly hurt performance--deserves careful interpretation:

Data diversity != data volume. The 1,110 volumes come from 624 patients. Augmentation creates variants of existing anatomies but cannot introduce new structural configurations. A model learning to detect RNFL thinning patterns needs diverse examples of *where* and *how* thinning manifests, not multiple exposure variants of the same thinning.

Small normal class. With only 263 healthy scans, even 6x augmentation produces only ~1,578 "normal" training examples--all derived from the same 263 optic nerve head anatomies. The model may not have enough variety to learn a robust "normal" template.

Augmentation noise floor. Synthetic noise and distortions, while physically motivated, may introduce subtle distribution shifts that the model treats as signal rather than nuisance variation.

Training dynamics. The 6x larger training set means each epoch takes much longer, and the effective number of unique gradient directions per epoch is diluted by near-duplicate volumes. This can slow convergence and encourage the model to memorize augmentation-specific patterns.

Consistency with literature. The original paper similarly found augmentation unhelpful (AUC dropped from 0.94 to 0.92), and their dataset is the same one used here. This suggests an intrinsic property of this particular dataset rather than a flaw in the augmentation approach.

6.2 Architecture Insights

The ResNet-Like model's consistent superiority suggests that:

- The classification task benefits from deeper feature hierarchies (enabled by residual connections) beyond what a 5-layer sequential stack can capture.
- The attention model's additional parameters may lead to overfitting on this small dataset, explaining its higher variance across configurations.

- All three architectures converge within a narrow AUC band (roughly 0.85-0.95), suggesting a performance ceiling imposed by dataset size and diversity.

6.3 Practical Implications

For deployment as a clinical screening tool:

- The **ResNet-Like model trained on original data** offers the best performance-to-complexity ratio.
- The model could serve as a triage system, flagging high-probability scans for ophthalmologist review.
- Inference is fast (single forward pass through a relatively shallow 3D CNN), making it suitable for point-of-care deployment.

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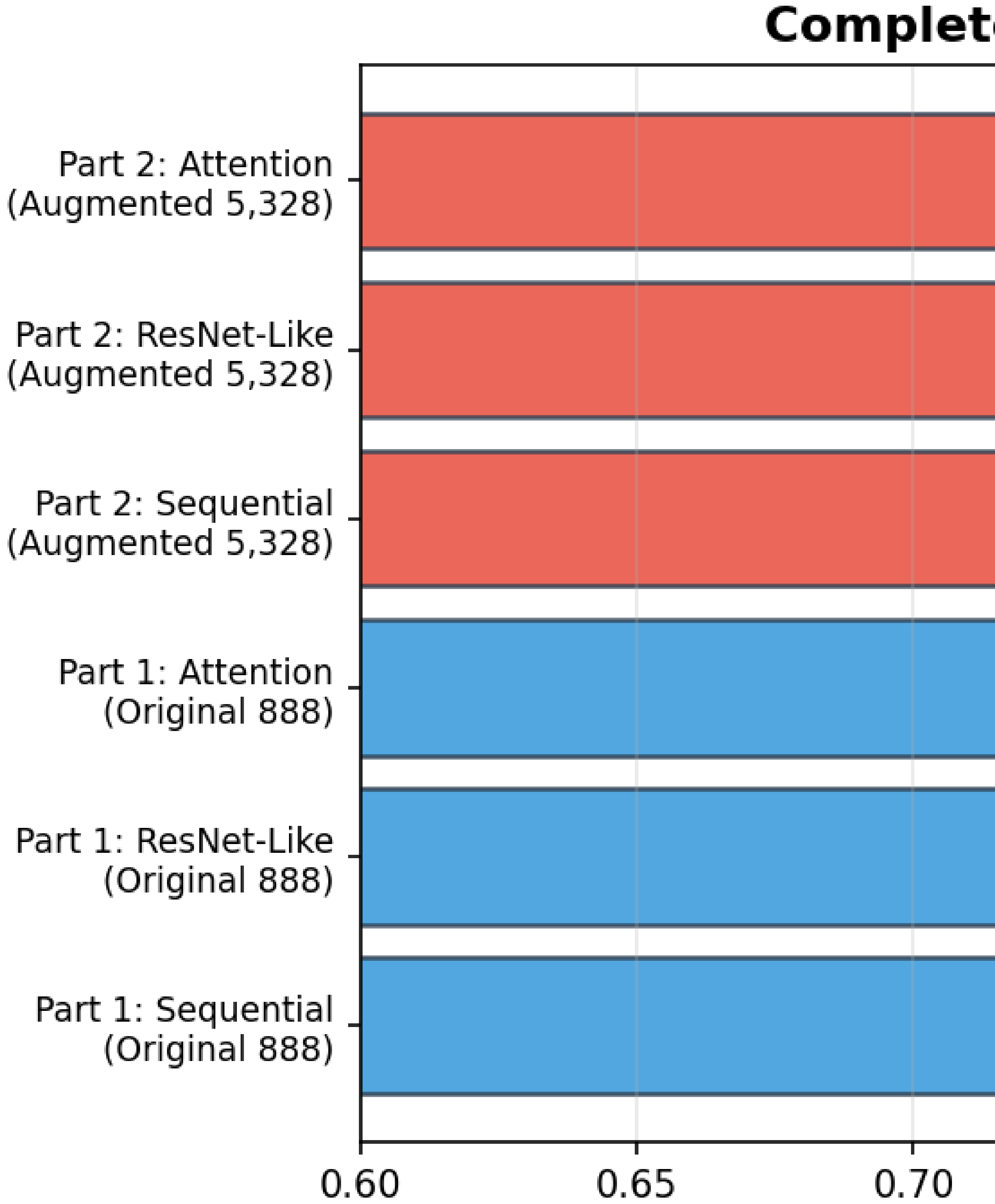
7. Conclusions



1. **Physics-informed augmentation** (gamma, noise, LPF, fan distortion) is a principled approach to expanding OCT datasets, but did not improve classification AUC for this specific dataset and task. This confirms the original paper's

finding.
2. **Architectural improvements** (residual connections, attention mechanisms) provide more benefit than data augmentation when the base dataset has limited patient diversity.
3. **The ResNet-Like architecture** remains the best model (AUC = 0.94 from Part 1), trained on original data without augmentation.
4. **The performance ceiling** (~0.94 AUC) appears to be a property of the dataset itself (624 patients, single institution, single scanner) rather than a modeling or data volume limitation.

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8. Future Directions

1. **5-Fold Cross-Validation**: Implement proper k-fold CV to provide confidence intervals and enable direct comparison with the paper's metrics.
2. **Multi-Center Data**: The clearest path to improved performance is incorporating OCT volumes from additional institutions and scanner types.
3. **Transfer Learning / Foundation Models**: Leverage pre-trained 3D medical imaging feature extractors.
4. **Grad-CAM / Explainability**: Generate class activation maps to validate that the model attends to clinically meaningful structures (RNFL, optic cup).
5. **Contrastive Self-Supervised Pretraining**: Use augmented pairs for learning scanner-invariant representations before supervised fine-tuning.
6. **2D Slice-Based Models**: Train lightweight 2D CNNs on extracted B-scans for faster inference and edge deployment.
7. **Web Application**: Deploy the best model via Gradio on Hugging Face Spaces for interactive screening demonstrations.
8. **Ensemble Methods**: Combine predictions across architectures and/or CV folds for more robust clinical predictions.

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9. Project Structure

PART_2/ DW EDA.ipynb # Data wrangling & exploratory analysis pps modeling.ipynb # Preprocessing (Part I) + Modeling (Part II) Capstone Three_Final_Report.md # This report Capstone Three_Presentation.pptx# Slide deck capstone_three_ *project_ideas.pdf capstone_three* _project_proposal.pdf [EDA visualization outputs]

octcv/ # Shared library code mdl_lib/ ? **init**.py # XVolSet, ModelEvaluator, utilities ? architectures.py # buildSequential, buildResNet, buildAttnNN ? augmentation.py # Augmentation transforms ? callbacks.py # LivePlot, EpochProgressBar arrViz.py # Array visualization utilities system_monitoring.py # GPU/RAM monitoring

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10. References

[1] Ishikawa, H. (2018). *OCT volumes for glaucoma detection* (Version 1.0.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.1481223

[2] Maetschke, S., Antony, B., Ishikawa, H., Wollstein, G., Schuman, J., & Garnavi, R. (2019). A feature agnostic approach for glaucoma detection in OCT volumes. *PLOS ONE*, 14(7), e0219126. https://doi.org/10.1371/journal.pone.0219126

[3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *CVPR*, 770-778.

[4] Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. *CVPR*, 7132-7141.

[5] Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, 6(1), 60.

Report written as part of OCTCV Capstone Three. Full code and reproducible notebooks available in the repository.